
Department of Civil Engineering, IIT Madras
CV2090 (Probability & Statistics for Civil Engineers)
Jul–Nov 2026

13 Aug, 2026

Tutorial-2

3:00–5:00 pm

Instructions:

1. Make assumptions wherever necessary, and *state* them explicitly.
 2. Tutorials are closed-phone, closed-tablet, and closed-LLM.
 3. You can bring books and notes, you are welcome to discuss with friends and TAs. However, your submission should be your own work.
 4. Submit handwritten solutions on A4 sheets, prepared during the tutorial hours.
 5. Submissions will not be accepted after the tutorial hours.
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Q1. Levee Overtopping Risk (Total Probability) A coastal flood defense levee protects an urban floodplain. During an annual hurricane season, the levee is subjected to three mutually exclusive storm surge severity states: Minor (S_1), Moderate (S_2), and Severe (S_3). Based on regional historical hydro-meteorological data, the annual probabilities of occurrence for these storm surge states are $\Pr(S_1) = 0.60$, $\Pr(S_2) = 0.30$, $\Pr(S_3) = 0.10$. Hydrologic modeling establishes that the conditional probability of the levee experiencing an overtopping failure (O) under each storm surge state is

$$\Pr(O \mid S_1) = 0.02, \quad \Pr(O \mid S_2) = 0.15, \quad \Pr(O \mid S_3) = 0.65.$$

Calculate the total annual probability that the levee experiences an overtopping failure, $\Pr(O)$.

Q2. Structural Flaw Detection via Non-Destructive Testing (Bayes' Rule)

A structural health monitoring protocol utilizes non-destructive ultrasonic testing (NDT) to inspect welded joints in steel bridge girders. In the regional bridge inventory, the baseline probability that a critical structural flaw (F) exists in a randomly selected weld is $\Pr(F) = 0.04$.

The NDT diagnostic instrument is imperfect:

- If a critical flaw is present, the test yields a positive detection signal (T^+) with a sensitivity (true positive rate) of

$$\Pr(T^+ \mid F) = 0.92.$$

- If a critical flaw is absent (F^c), the test correctly yields a negative signal (T^-) with a specificity (true negative rate) of

$$\Pr(T^- \mid F^c) = 0.95.$$

An inspector performs a single NDT scan on a critical girder weld, and the instrument indicates a positive detection signal (T^+). Calculate:

- (a) The total probability of receiving a positive test result across the bridge inventory, $\Pr(T^+)$.

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- (b) The updated posterior probability that the weld actually contains a critical structural flaw given the positive test result, $\Pr(F \mid T^+)$.

Q3. Concrete Batch Quality Control and Project Delays (Discrete Random Variables) A highway construction project receives a daily shipment of $n = 4$ independent ready-mix concrete batches. Due to variability in aggregate moisture and batch plant mixing, each batch has a constant probability $p = 0.10$ of failing compressive strength quality control testing. Let K denote the number of non-conforming concrete batches identified in a daily delivery.

The structural site engineer maps the number of non-conforming batches K to an operational schedule delay X (in days) according to the following physical response function:

- If $K = 0$ failed batches, $X = 0$ days delay.
- If $K = 1$ failed batch, $X = 1$ day delay (for rapid field re-testing).
- If $K = 2$ failed batches, $X = 3$ days delay (for core extraction and laboratory curing).
- If $K \geq 3$ failed batches, $X = 7$ days delay (for full structural demounting and batch rejection).

Determine:

- (a) The discrete probability mass function (PMF) of the project delay random variable X , tabulated as $\Pr(X = x)$ for all valid support values of X .
- (b) The expected project delay $\mathbb{E}[X]$ in days.

Q4. Traffic Sensor Network Diagnostic Modeling (PMF Construction & Parameter Derivation) An intelligent transportation system monitors traffic density across a critical arterial corridor using a four-node sensor array. System diagnostics record an operational failure state index $X \in \{0, 1, 2, 3, 4\}$, where X represents the severity index of active sensor failures across the array. Reliability engineers propose a discrete mathematical model for the probability mass function of X :

$$\Pr(X = x) = c \cdot (x^2 + 1), \text{ for } x \in \{0, 1, 2, 3, 4\},$$

where c is an unknown real-valued normalization constant. Calculate:

- (a) The exact numerical value of the normalization constant c required to establish a mathematically valid PMF.
- (b) The complete probability mass function $\Pr(X = x)$ expressed as exact fractions and decimal equivalents.

Q5. Pavement Fatigue Failure Reliability (Binomial Distribution) A newly constructed interstate highway segment is divided into $n = 8$ structurally independent sub-segments subjected to heavy commercial axle loadings. Empirical pavement degradation models indicate that each individual sub-segment has a constant probability $p = 0.15$ of developing severe fatigue cracking within a 5-year design life. Assuming sub-segment failures occur independently, calculate:

- (a) The probability that exactly 0, exactly 1, and exactly 2 sub-segments experience fatigue cracking within the 5-year window.
- (b) The probability that at least 2 sub-segments experience fatigue cracking within the 5-year design life, $\Pr(X \geq 2)$, triggering major rehabilitation work.

Q6. Derivation of the Poisson Distribution Suppose that random events (e.g., phone calls arriving at a call center or vehicles arriving at a toll booth) occur randomly over a fixed time interval. Divide the interval into n very small sub-intervals such that the probability of exactly one event occurring in each sub-interval is p , and the probability of two or more events occurring in a sub-interval is negligible. Assume events occurring in different sub-intervals are independent.

- (a) Show that the total number of events, X , occurring during the entire time interval follows a Binomial distribution with parameters n and p , i.e.,

$$\Pr(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n.$$

- (b) Suppose that

$$n \rightarrow \infty, \quad p \rightarrow 0, \quad \text{while } np = \lambda$$

remains constant. Show mathematically that

$$\lim_{n \rightarrow \infty} \binom{n}{x} p^x (1 - p)^{n-x} = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

Hence, derive the probability mass function of the Poisson distribution.